

## The Periodic Table of Elements

The periodic table organizes all known chemical elements by atomic number, electron configuration, and chemical properties.

### Key Element Groups:

- Group 1 (Alkali metals): Li, Na, K
- Group 17 (Halogens): F, Cl, Br, I
- Group 18 (Noble gases): He, Ne, Ar

### Chemical Bonding

1. Ionic bonds: Transfer of electrons (e.g., NaCl)
2. Covalent bonds: Sharing of electrons (e.g., H<sub>2</sub>O)
3. Metallic bonds: Sea of shared electrons in metals